

AUTO COIN-SORTER (₹45)



Got a lot of loose change and just want to start using it? What turns you off is probably having to sort them by denomination, right? Let that worry you no more...

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MATERIALS NEEDED

- Cardboard: Free
- 9V DC motor: ₹15
- 9V battery: ₹15
- Coins: As per value
- Wires: ₹5
- N-type transistor: ₹5
- 10K linear potentiometer: ₹10
- Toggle switch: ₹5

If you happen to magically be in possession of a lot of currency coins by the end of the month then you'll probably have a harrowing time sorting them out before you can even count them. This nifty DIY can take care of the sorting part with ease. Well, until it could do that until all Indian Currency coins were distinct. Now with the new Rs.1 being of the same size as the older Rs.2 coins, it can even confuse machines let alone humans. We'll try anyways.



STEP 01 The basic principle behind the coin sorter is that currency coins of different values come in different sizes. We're simply cutting slots on a conveyor such that only the same value coins fall into the same tray.



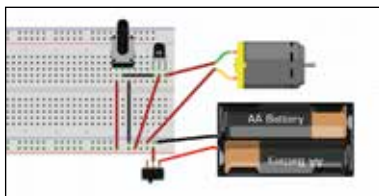
STEP 02 We begin by building a tray. For this sorter we are going to be sorting the newer Rs. 1, 2, 5 and 10 denomination coins. We simply traced out the dimensions of a tray and cut it out. The only dimension that matters here is that each tray be 5 cms wide.



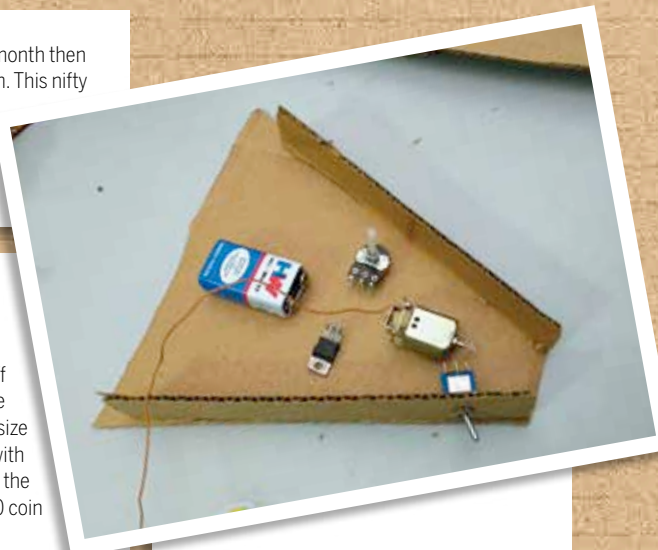
STEP 03 Next, we build the sorter by tracing the sizes of the coins. We only need to cut 3 slots since the 4th size coin will drop off the end. Please make sure you are cutting in the increasing order of size as the sorter descends. So start with the Rs. 1 coin, then the Rs. 5, then the Rs. 2 coin and finally, let the Rs. 10 coin fall off.



STEP 04 This is the automatic part. The coins are automatically fed into the sorter by the feed tray which is constantly vibrating. The feed tray is wide on one end and tapers towards the other end until it is just wider than the Rs. 10 coin.



STEP 05 The tray is violently shaken by a biased motor. If you can't get a biased motor, simply add a glob of hot glue to the shaft of the



motor like we did and then follow this circuit diagram. A more detailed version can be found on <https://geek.digit.in>

STEP 06 Attach the biased motor circuit to the underside of the feeder tray. And then punch a hole on the side of the feeder wall to install the switch that turns the feeder system on. Secure everything with hot glue.



STEP 07 Now put everything together. Glue the feeder tray

STEP 08 Switch it on and start sorting your coins.